DATA130024.01 – Fundamentals of Probability Theory

12:45 -14:45 pm, December 26, 2022, Monday

Final Exam (Closed Book)

Problem 1) (Subtotal: 20 marks)

Let X and Y be continuous random variables such that (X, Y) must belong to the square in the xy-plane containing all points (x, y) that satisfy $2 \leq x \leq 4$ and $1 \leq y \leq 3$. Suppose that the joint cumulative distribution function of X and Y at every point (x, y) in this square is

$$F(x, y) = \frac{1}{e^c}(x - a)(y - b)e^{x+2y},$$

where a, b and c are unspecified constants.

(i) Find the values of a, b and c. Please justify your answers. (5 marks)

(ii) Show if X and Y are independent. (7 marks)

(iii) Determine the value of $\frac{\mathbf{E}[XY]}{\mathbf{E}[Y]}$. (8 marks)

Problem 2) (Subtotal 20 marks)

For every $\lambda > 0$, let X_λ denote the Poisson random variable with parameter λ, define the random variable $Y_\lambda = \frac{X_\lambda - \lambda}{\sqrt{\lambda}}$, and let $\psi_\lambda(t)$ be the moment generating function of Y_λ.

(i) Determine $\psi_\lambda(t)$ in terms of λ and t. (5 marks)

(ii) Determine $\lim_{\lambda \to \infty} \psi_\lambda(t)$. (5 marks)

(iii) Using the limit you computed in part (ii), explain in **as much details as possible** why and how the Poisson distribution can be approximated by a normal distribution. Please include details on conditions for which the approximation is valid. For example, what are the parameters of the normal distribution that approximates a Poisson distribution with parameter λ? Please also include details explaining why such an approximation is justified. (10 marks)

Problem 3) (Subtotal: 20 marks)

A group of N people put their neckties into the box. After shaking the box to mix up neckties, each person randomly picks tie from the box. Let X be the number of people that select their own tie.

(i) Show that $E[X] = 1$

(ii) What is $Var(X)$?

Problem 4) (Subtotal: 20 marks)

A random sequence X_n converges in the mean square sense to X if

$$E[|X_n - X|^2] \to 0, n \to \infty.$$

A random sequence X_n converges in probability to X if for every $\epsilon > 0$,

$$P(|X_n - X| > \epsilon) \to 0, n \to \infty.$$

Show that convergence in the mean square sense implies the convergence in probability. (Hint: you may use Chebyshev inequality.)

Problem 5) (Subtotal: 10 marks)
Suppose a jar contains $2N$ cards, two of them marked 1, two marked 2, and so on. Draw out m cards at random. What is the expected number of pairs that still remain in the jar? (Hint: you may define a bernoulli random variable X_i that takes on value 1 when i-th pair remains in the jar and 0 otherwise.)

Problem 6) (Subtotal: 10 marks)
A random variable R is called to have Rayleigh distribution if its density function is given by

$$f_R(r) = \frac{r}{\sigma^2} e^{-\frac{r^2}{2\sigma^2}}.$$

Find out the expected value of R.

Please write your name and student No. on each page of your answer sheet!